

PROJECT DESCRIPTION

Enable MFA for IAM User and Federated User Using Authenticator App

Project Domain: Cloud Computing / Amazon Web Services (AWS)

Project Type: Academic Project

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1. Introduction

Cloud computing provides flexible access to computing resources, but secure identity and access management is essential because cloud accounts can control important services and data. This project demonstrates the use of Multi-Factor Authentication (MFA) in Amazon Web Services (AWS) for both an IAM user and a federated user. The authentication process uses an authenticator application to provide an additional verification factor beyond the normal password.

The project is focused on practical configuration and verification. In the IAM environment, a test user is configured with AWS permissions and virtual MFA. The project also demonstrates a FederatedUser in AWS IAM Identity Center, where an authenticator app is registered as an MFA device. The supplied synopsis shows the AWS console, IAM users, user permissions, MFA status, IAM Identity Center, federated user details, and sign-in screens used during the implementation.

2. Objectives of the Project

The main objectives of the project are:

- To understand AWS Identity and Access Management and the role of authentication in cloud security.
- To create and configure an IAM test user for secure AWS console access.
- To attach AmazonEC2FullAccess and AmazonS3ReadOnlyAccess policies to the IAM user.
- To configure a virtual MFA device using an authenticator application.
- To understand the difference between a regular IAM user and a federated user managed through IAM Identity Center.

- To register an authenticator app as an MFA device for the federated user and observe the sign-in process.
- To verify that MFA configuration is visible in the AWS security and user-management interfaces.

3. AWS IAM User MFA Implementation

The first part of the project deals with an IAM user. The AWS IAM dashboard provides access to users, groups, roles, policies, identity providers, account settings and security-related options. In the supplied project material, the IAM users list contains AdminUser, DeveloperUser and MFATestUser. The MFATestUser entry shows "Virtual" in the MFA column, indicating that virtual MFA has been enabled.

The test user's summary page shows console access enabled with MFA. The Permissions tab contains two attached policies: AmazonEC2FullAccess and AmazonS3ReadOnlyAccess. These permissions provide the user with full access to Amazon EC2 and read-only access to Amazon S3 for the purpose of the academic demonstration.

The implementation can be understood through the following sequence:

- Open the AWS Management Console and navigate to IAM.
- Open the IAM Users section and select the test user.
- Review the user's console-access and security settings.
- Attach the required policies: AmazonEC2FullAccess and AmazonS3ReadOnlyAccess.
- In the Multi-factor authentication section, assign a virtual MFA device.
- Use an authenticator application to scan the QR code and generate verification codes.
- Complete MFA registration and verify that the user shows a virtual MFA device.
- Test console sign-in using the IAM username, password and MFA verification.

4. IAM User Sign-in and Verification

The supplied screenshots show the IAM user sign-in form containing the Account ID or Alias, IAM username and Password fields. After the initial credentials are supplied, MFA is used as an additional authentication step. The IAM user security page also provides controls to remove, resynchronize or assign a new virtual MFA device. This confirms that MFA is associated with the user account rather than being only a visual setting on the dashboard.

The project therefore demonstrates the basic security principle of combining something the user knows, such as a password, with a separate authentication factor generated by an authenticator application. This additional factor makes unauthorized access more difficult when only the password is known.

5. Federated User MFA Implementation

The second part of the project focuses on a federated user through AWS IAM Identity Center. The supplied material shows IAM Identity Center as the service used to manage workforce access to AWS accounts and cloud applications. Its interface includes Users, Groups, Settings, Multi-account permissions and Application assignments.

A user named FederatedUser, with display name Ojas Saraswat, is shown in the Users section. The user status is Enabled, the user was created manually, and one MFA device is attached. The user details page contains Profile, Groups, AWS accounts, Applications, MFA devices and Active sessions tabs. The MFA devices tab shows one registered authenticator app, created on August 24, 2026.

The federated-user workflow demonstrated in the project is:

- Open AWS IAM Identity Center and access the Users section.
- Select the FederatedUser account.
- Review the user's profile and MFA device information.
- Register an authenticator application as an MFA device.
- Complete the required verification using the authenticator-generated code.
- Proceed to the AWS sign-in page using the federated-user credentials.
- Verify that the authentication flow recognizes the configured security factor.

6. Comparison of IAM User and Federated User

Feature	IAM User	Federated User
Management area	AWS IAM	AWS IAM Identity Center
Example shown	MFATestUser	FederatedUser
MFA	Virtual MFA device	Authenticator app MFA device
Permissions/access	IAM policies attached to user	Workforce/application/AWS account access
Sign-in	IAM user sign-in page	Identity Center/federated sign-in flow

7. Security Significance

MFA is important in cloud environments because a password alone can be exposed through guessing, reuse or other credential-compromise scenarios. By requiring an additional verification factor, the authentication process adds another security layer. In this project, the authenticator application is used to generate the verification factor for the configured accounts.

The project also highlights that access control and authentication are related but different concepts. IAM policies determine what an authenticated IAM user can access, while MFA strengthens the process used to establish the user's identity. The supplied IAM dashboard additionally displays security recommendations related to enabling MFA for the root user and managing root access keys.

8. Results and Observations

The implementation successfully demonstrates the configuration of MFA for the IAM test user and the registration of an authenticator app for the federated user. The IAM user interface displays a virtual MFA device, while the FederatedUser details show one MFA device and an authenticator app registration. The sign-in screens provide the practical verification stage of the project.

The project also provides hands-on understanding of AWS IAM, IAM Identity Center, user management, permissions, console access and multi-factor authentication. The supplied material records the AWS console configuration and the corresponding user and authentication screens used during the demonstration.

9. Conclusion

This project demonstrates how AWS authentication can be strengthened by enabling Multi-Factor Authentication with an authenticator application. The first implementation uses an IAM user with specified AWS permissions and a virtual MFA device. The second implementation uses AWS IAM Identity Center with a FederatedUser and a registered authenticator app. Together, these demonstrations provide a practical understanding of identity security in cloud computing.

The project concludes that MFA is an important security layer for protecting cloud accounts and reducing the risk associated with password-only authentication. It also gives students practical exposure to AWS IAM and IAM Identity Center, making the concepts of authentication, authorization and federated access easier to understand through real console-based configuration.

10. Future Scope

The project can be extended by studying MFA enforcement for multiple users, group-based permission management, stronger access policies, temporary credentials, application assignments and centralized workforce access across multiple AWS accounts. Additional testing can also be performed to compare different authentication and access-management configurations.